Mark Hanna

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Technical Skills

Languages: Java, C, C++, C#, Python, JavaScript, SQL, MATLAB, Visual Basic, HTML, CSS
Back End & Databases: Node, Express, PostgreSQL, REST APIs, Query Optimization, Input Validation, Logging Middleware, Cron Jobs, Queue Workers
Tools & Platforms: Git, GitHub, Docker, Postman, Unity, Quartus Prime, SolidWorks, OnShape, Excel with VBA, Visual Studio Code
Hardware & Embedded: VHDL, FPGA, STM32 MC, Photonics Circuits (PIC), CAN Protocol, Simulink Simulations, Oscilloscopes
Micro controller & Firmware: STM32CubeIDE, Embedded C++, Sensor Integration, Real Time Data Logging

Experience

Back End Developer May 2025 - Sep. 2025
RM Health Supplies *Mississauga, ON*

- Engineered REST APIs using **Node, Express,** and **SQL** that supported **5 internal tools**.
- Implemented indexing and query restructuring in **PostgreSQL**, raised throughput by **25 percent**.
- Diagnosed a **memory leak** in a long-running service, reduced crash frequency from **4 per week to 0**.
- Deployed background jobs with **cron** and **queue workers**, cut manual admin tasks by **60 percent**.
- Added **input validation** and **logging middleware**, reduced error rates by **15 percent**.

Computer Technician Sep. 2023 - Dec. 2024
Technoid Gaming Computers *Mississauga, ON*

- Produced assembly instructions and diagrams that reduced build time by **20 percent.**
- Ensured **safe workflow** and accurate documentation for all builds and repairs.

Projects

Dobby's Night Out Apr. 2026
Team Project (5 members) *C#, Unity 6, Git*

- Built a **5-level** RPG in **Unity 6** with a team of **5** in **3 weeks**, merging work via **Git** pull requests.
- Prototyped the **Cat Smack** ability in **C#**, with hit detection, a cooldown and an **IDamageable** interface.

4 Bit ALU on FPGA Jun. 2025
Personal Project *VHDL, FPGA, Hardware Verification, RTL Design*

- Engineered a 4-bit ALU with **28 logic gates** and a **4-bit ripple-carry adder,** validated at **50 MHz**.
- Simulated worst case propagation delay of **18 ns**, matched FPGA testing within **5 percent**.
- Used **2 percent** of logic elements on **Intel MAX10**, confirmed stable output on LEDs and 7-segment displays.

Fall Detection System w/ STM32 Nov. 2024
Personal Project *C/C++, Micro-controllers, SolidWorks, STM32CubeIDE*

- Implemented accelerometer sampling at **200 Hz** using **C++** and **STM32**, with threshold-based logic.
- Delivered GSM alerts in under **3 s** with a **92 percent** true-positive detection rate.

Extra Curricular

Western Baja Racing Sep. 2025 - Present
Western University *London, ON*

- Implemented **UART-based data acquisition** on **STM32**, processed **500 plus samples per second**.
- Optimized embedded filtering logic, lowered sensor read errors by **30 percent** during vibration tests.
- Helped deploy a **Python dashboard** with real-time plots, cut analysis time by **40 percent**.

Education

Western University London, ON
Bachelor of Engineering Sciences - BESc, Software Engineering *Expected Graduation: May 2029*